Class XI Physics — Unit Test

Time: 1 hour 30 minutes Maximum Marks: 25

General Instructions:

1. All questions are compulsory.

2. Marks for each question are indicated against it.

SECTION A

1. Which physical quantity is a vector quantity? 1

(a) speed (b) velocity (c) distance (d) mass

2. The slope of a velocity-time graph represents which physical quantity? 1

(a) displacement (b) velocity (c) acceleration (d) force

3. A ball is thrown vertically upward with a speed of 20 m/s. Taking

g = 10 m/s2, calculate its maximum height. 1

SECTION B

4. State Newton's second law of motion and derive the relation F = ma. 3

5. Define the work done by a constant force and state its SI unit. 3

SECTION C

6. A projectile is fired from the ground with a speed of 20 m/s at an angle of 30 degrees to the horizontal. Calculate:

(a) the time of flight. 2

(b) the maximum height reached. 3

(c) the horizontal range. 3

7. A block of mass 5 kg is pulled along a rough horizontal surface by a horizontal force of 25 N. The coefficient of kinetic friction is 0.3.

Take g = 10 m/s2.

(a) Draw the free-body diagram of the block. 2

(b) Calculate the frictional force acting on the block. 3

(c) Calculate the net acceleration of the block. 3